

Eman Chaudhary

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EDUCATION

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelor of Engineering in Software Engineering

Sep 2022 – May 2026

TECHNICAL SKILLS

LLMs & Generative AI: LLM Inference, Prompt Engineering, Fine-Tuning (LoRA/PEFT), RAG Pipelines, Hybrid Retrieval (Dense + BM25), Structured Data Extraction, Agentic Systems, Tool Use, LangGraph, LangChain, LlamaIndex, Hugging Face, Ollama, sentence-transformers, Diffusion Models

Backend & APIs: FastAPI, Django, Node.js, Express.js, REST APIs, WebSockets, Celery, Redis, Async Pipelines

ML & Predictive Analytics: PyTorch, TensorFlow, LSTM, Random Forest, GBM, Anomaly Detection, CNN

MLOps & Infrastructure: Docker, Databricks SQL, Delta Tables, MLflow Experiment Tracking, Vector Databases (Qdrant, FAISS)

AI Safety & Evaluation: Guardrails & Evaluation Frameworks, Prompt-Injection Detection, Confidence Gating, PII Anonymization, Hallucination Defense, Audit Logging

Languages: Python, JavaScript/TypeScript, SQL

PROJECTS

AI-Driven Customer Service Assistant for NUST Bank (RAG System)

Jan 2026 – Feb 2026

Gemma-3-4B, PEFT/LoRA, Qdrant, sentence-transformers, BM25, Presidio, Celery, FastAPI, Redis, Docker

- Built an end-to-end RAG pipeline with hybrid dense + BM25 sparse retrieval and a cross-encoder reranker; reduced irrelevant context chunks by an estimated ~20% compared to dense-only retrieval on a 100-query test set.
- Fine-tuned Gemma-3-4B via PEFT/LoRA (rank=8, alpha=16) on domain-specific banking FAQ and policy Q&A pairs to enforce brand tone and prevent out-of-domain hallucinations.
- Added a multi-layer safety stack: DeBERTa-based prompt-injection classifier, post-generation confidence gating, and Presidio + spaCy PII anonymization, caught the 95% of injection attempts in internal testing (n= 40 prompts). Deployed async FastAPI backend with Qdrant vector store

Industrial Predictive Maintenance System (NASA C-MAPSS)

June 2026

Python, scikit-learn, PyTorch (LSTM), MLflow, FastAPI, Streamlit, Ollama, Databricks SQL, Docker Compose

- ML Pipeline & Evaluation: Tracked 20+ experiment runs in MLflow, selecting a GBM (10.31 NASA score) over LSTM baselines for production RUL prediction.
- Risk Classification & Alerting: Trained failure risk and anomaly models, routing maintenance priorities to Databricks Delta tables for the critical engines.
- Dashboard: Containerized the full stack via Docker and integrated Llama 3.2 via FastAPI to generate plain-language maintenance briefings.

SmartSketch: AI-Powered Forensic Facial Composite Platform

Aug 2025 – May 2026

LangGraph, Django, FastAPI, Hugging Face Diffusers, SDXL, ControlNet, Qwen 2.5, OpenAI CLIP, FaceNet, GFPGAN

- Co-developed in a team of 3, owned the core multi agent orchestration and machine learning workflows.
- Built a 4-stage LangGraph multi-agent pipeline (analyze → route → artist → verify) with checkpointed state and Pydantic-validated outputs; end-to-end composite generation
- Implemented a closed-loop evaluation stage using CLIP cosine similarity and FaceNet identity scoring with a self-hosted Qwen2.5-VL critic; auto-revision loop with average CLIP scores increasing from 0.61 to 0.65
- Designed a three-tier system (React SPA → Django REST with JWT/Postgres/audit hooks → Modal GPU workers) with an audit trail

EXPERIENCE

Machine Learning Research Project (DNA Damage & Imaging)

Sept 2025 – Nov 2025

NASA OSDR AIML Analysis Working Groups

Remote (United States)

- Collaborated with AIML scientists on classifying DNA damage patterns from microscopy images; improved F1 macro on a severely imbalanced 6-class dataset.

- Used an attention-based CNN and an optimized ArrayRecord + Grain data loading pipeline that reduced per-epoch data loading time by $\sim 14\%$.

Machine Learning Research Intern

Jun 2025 – Sep 2025

SINES – EmbedAIoT Lab, NUST

Islamabad, Pakistan

- Designed lightweight **CNN and CNN–Transformer** architectures (101–745 parameters) for 5G mmWave channel estimation; best model achieved **NMSE of -11.9 dB**, outperforming LS and MMSE baselines, with < 1 ms inference on edge hardware via INT8 quantization and structured pruning.
- Results published at ICODT(IEEE): <https://ieeexplore.ieee.org/document/11360709>

Software Development Intern

Jun 2024 – Sep 2024

Apna-Wifi (NSTP, NUST)

Islamabad, Pakistan

- Built Python web scraping pipelines (Selenium) to extract and ingest around 5,000 course records into a MySQL database.
- Developed a Django-based recommendation engine utilizing BERT and DistilBERT semantic embeddings, achieving a **78% Hit Rate@5** on an evaluation set.
- Collaborated closely within an agile startup environment under the supervision of the technical lead to build software matching client specifications.

LANGUAGES

English: Fluent **Urdu:** Fluent **Arabic:** Basic **German:** Basic